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Unlabeled kratom alkaloids detected in products marketed as kava

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ABSTRACT

Introduction: In October 2025, the Alabama Poison Information Center and local addiction medicine providers identified a cluster of cases involving liquid products marketed as containing kava associated with opioid-like withdrawal symptoms. Kava is not typically associated with physical dependence, raising concern for opioid-active substances.

Methods: Five unique retail products were submitted for laboratory analysis. Three underwent quantitative ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) testing for kavalactones, kratom alkaloids, and related semisynthetic derivatives, with additional screening for 260 substances. Two products underwent qualitative gas chromatography–mass spectrometry (GC–MS) testing for mitragynine and kavalactones.

Results: Qualitative GC–MS detected both mitragynine and kavalactones in one product, while another contained mitragynine with no detectable kavalactones. Quantitative UPLC–MS/MS identified both kavalactones and kratom alkaloids in three products, with mitragynine pseudoindoxyl predominating at concentrations of 0.16–0.21 mg per bottle.

Discussion: Detection of kratom alkaloids and related semisynthetic derivatives in products labeled as kava may explain the observed opioid-like withdrawal. Mitragynine pseudoindoxyl, which has greater opioid activity than mitragynine and 7-hydroxymitragynine, was identified as the predominant kratom-related compound.

Conclusion: Unrecognized exposure to opioid-active kratom compounds may occur through products marketed as kava, highlighting the need for improved surveillance and increased awareness.

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Introduction

Adulteration and mislabeling of botanical and dietary supplements, both intentional and unintentional, are well described in the literature, including reports of undisclosed ingredients and the addition of pharmacologically active compounds to enhance effects [1]. Limited regulatory oversight and routine testing make it difficult for consumers and providers to determine actual exposure.

In October 2025, the Alabama Poison Information Center, in collaboration with local addiction medicine providers, identified a cluster of cases involving liquid products labeled as containing *Piper methysticum* (kava) purchased from gas stations, convenience stores, and specialty shops. Calls involved patients aged 17–54 years reporting daily use with escalating doses and withdrawal symptoms upon cessation, prompting presentation to health care facilities. Several patients described symptoms partially resembling opioid withdrawal, and some experienced improvement after buprenorphine initiation.

Kavalactones are not typically associated with physical dependence; however, rare withdrawal-like symptoms have been described [2]. These products additionally claimed to contain *Uncaria tomentosa* (cat's claw) and/or *Picralima nitida* (akuamma), and

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neither substance has been clearly associated with clinically significant withdrawal. Published data on akuamma-related dependence or withdrawal are limited, with recent survey data suggesting a low risk of akuamma-related use disorder [3]. Given discordance between reported exposures and clinical presentation, and the absence of similar reports in the literature, informal surveillance of publicly available discussions suggested these products might contain kratom or related high-potency semisynthetic derivatives, such as 7-hydromitragynine and/or mitragynine pseudoindoxyl. Prior analyses of kratom-labeled products have demonstrated 7-hydroxymitragynine at concentrations unlikely to reflect native leaf composition, suggesting intentional addition or mislabeling [4,5]. Furthermore, these substances are not commonly included in standard toxicologic testing, which may complicate identification and delay treatment. We sought to evaluate these commercially available products for evidence of kratom alkaloids, related semisynthetic derivatives, and other illicit substances given the concern for unrecognized opioid exposure and potential legal and public health implications.

Methods

Selected products represented those most frequently referenced by patients and clinicians and available at local retail locations. Samples were unopened, maintained in original packaging, and submitted for laboratory analysis. Two independent analytical laboratories were utilized based on differences in testing capabilities and availability of methods. Three products underwent quantitative testing for kavalactones, kratom alkaloids, and kratom-related derivatives using ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) methods and were screened for 260 additional substances using the panel and approach described in the Sharma et al.'s protocol [6].

Two additional products underwent qualitative gas chromatography–mass spectrometry (GC–MS) testing, limited to detection of kavalactones and mitragynine based on available standards. This analysis was not quantitative and was not designed to comprehensively identify kratom alkaloids or related semisynthetic derivatives but allowed for rapid on-site analysis.

Results

None of the five products claimed to contain kratom alkaloids or related compounds. Of the five products analyzed, two underwent qualitative GC–MS analysis, which was limited to detecting mitragynine and kavalactones. One product contained detectable levels of both mitragynine and kavalactones. The other detected only mitragynine, with no detectable kavalactones despite the label claim. Three products underwent quantitative UPLC–MS/MS testing, which identified kavalactones as well as mitragynine, 7-hydroxymitragynine, and mitragynine pseudoindoxyl. The predominant kratom-related substance in UPLC-MS/MS-tested products was mitragynine pseudoindoxyl, with amounts ranging from 0.16 to 0.21 mg per bottle (Table 1). Targeted UPLC–MS/MS screening did not detect prescription or illicit opioids.

Discussion

Kratom alkaloids and related derivatives, including mitragynine pseudoindoxyl, were identified in products marketed as kava without disclosure on labeling, raising concern for intentional misrepresentation of widely available products. Kava products are marketed as calming supplements, and consumers may not anticipate opioid-active compounds. Such exposure may lead to escalating use, dependence, and withdrawal, and present diagnostic and management challenges for clinicians when patients report sole use of kava and have

Table 1. UPLC-MS/MS quantitative analysis of kava-labeled products.

Product name	Happy Hour Shot 777, mixed berry	Happy Hour Shot, BlueBurstPop	Boujee Bliss, strawberry banana
Label claim	Kavalactone extract 125 mg Akuamma 500 mg Cat's claw 750 mg	Kavalactone extract 125 mg Akuamma 500 mg Cat's claw 750 mg	*Piper methysticum* extract *Uncaria tomentosa* extract
Amount in each bottle (fl oz)	1.0	1.0	0.5
Total kratom alkaloids (µg/bottle)*	263 ± 9.6	258.5 ± 12.0	221.3 ± 4.0
Mitragynine (µg/bottle)	38.7 ± 1.9	34.3 ± 0.5	38.5 ± 1.1
7-hydroxymitragynine (µg/bottle)	10.5 ± 0.7	11.5 ± 0.7	3.41 ± 0.0
Mitragynine pseudoindoxyl (µg/bottle)	207.2 ± 6.9	206.5 ± 10.8	163.4 ± 2.6
Total kavalactones (mg/bottle)†	167.5 ± 1.2	158.1 ± 0.3	92.8 ± 0.1

Abbreviations: UPLC-MS/MS = ultra-performance liquid chromatography–tandem mass spectrometry, mg = milligram, fl oz = fluid ounce, µg = microgram.
*Kratom alkaloids tested for include: mitragynine, speciogynine, speciociliatine, mitraciliatine, 7-hydroxymitragynine, mitragynine pseudoindoxyl, corynantheidine, paynantheine, corynoxine, corynoxine B, isopaynantheine, mitraphylline, and speciofoline.
†Kavalactones tested for include: kavain, dihydrokavain, methysticin, dihydromethysticin, yangonin, desmethoxyyangonin, flavokavain A, and flavokavain B.

negative routine toxicology screens, potentially delaying recognition and treatment. These findings highlight the need for improved product transparency, public health messaging, clinician awareness, surveillance, and regulatory oversight to inform consumers of the potential for unlabeled opioid-active compounds in these products, particularly given the Schedule I status of kratom and related derivatives in many states.

Mitragynine is the primary alkaloid in *Mitragyna speciosa* (kratom), whereas 7-hydroxymitragynine is the primary active metabolite of mitragynine that can also be produced synthetically [7]. Mitragynine pseudoindoxyl is a related semisynthetic derivative formed through synthetic processes or *in vivo* from 7-hydroxymitragynine [7]. In preclinical studies, mitragynine pseudoindoxyl demonstrated substantially greater μ-opioid receptor potency than mitragynine and 7-hydroxymitragynine *in vitro*, and products containing higher concentrations of these compounds raise concern for increased opioid-like effects and risk of dependence and withdrawal [8]. Detection of a potent opioid receptor agonist provides a plausible explanation for the observed clinical presentations. Although the clinical significance of the detected concentrations remains uncertain, the predominance of mitragynine pseudoindoxyl supports concern for intentional adulteration or mislabeling.

Co-formulated kava and kratom products are available online and in states without kratom restrictions, and prior literature has evaluated this combination; however, the kratom studied was plant-derived and did not contain semisynthetic derivatives [9]. Additionally, the effects of kratom alkaloids and kavalactones may be altered through cytochrome P450-mediated pharmacokinetic interactions, as these substances share metabolic pathways and enzyme inhibition [10].

No additional substances of clinical concern, including prescription or illicit opioids, were identified; however, cat's claw and akuamma alkaloids were not included in the analytical panel, representing a limitation of this targeted approach. Additionally, only a limited number of products were tested, and selection was based on reported exposures rather than systematic market sampling, which may limit generalizability.

Conclusion

Kratom alkaloids and related semisynthetic derivatives were identified in products marketed as kava despite no disclosure on product labeling, raising concern for intentional adulteration or mislabeling and unrecognized exposure to opioid-active compounds in widely available retail products. Clinicians should consider exposure to kratom alkaloids or related compounds through products labeled as kava in patients presenting with opioid-like withdrawal, particularly when opioid use is denied. Continued surveillance, laboratory testing, and information sharing among poison centers, clinicians, law enforcement, and public health agencies are warranted to detect and address improperly labeled products marketed as "kava."

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Disclosure statement

CRM serves as an expert witness in kratom lawsuits for plaintiffs and defendants. The authors have no other competing interests to declare.

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Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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